

Bigya Tuladhar

Senior Full-Stack Developer · AI & RAG Systems · Cloud & Serverless Architecture

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SUMMARY

Senior Full-Stack Developer with 8+ years of experience building and scaling cloud applications across the full development life cycle. Works extensively with AI — building AI agents, RAG (Retrieval-Augmented Generation) systems, and AI-powered assistants — alongside AWS serverless architecture, real-time interactive systems, low-latency video streaming, and CI/CD automation. Experienced in building moderator and audience tools for large-scale live enterprise events, leading development teams, and improving system performance, scalability, and cost. Comfortable owning features end to end, from architecture and backend APIs to polished, responsive frontends.

EXPERIENCE

Senior Full-Stack Developer / Tech Lead

Feb 2022 – Present · Hong Kong

Nova Dynamic Media

Virtual-event company delivering live production, webcasting, and audience-engagement solutions.

- Built an **AI copilot for live-event moderators** using a **Retrieval-Augmented Generation (RAG)** system (on AWS Bedrock), indexing event content into a vector knowledge base so the AI returns accurate, grounded answers, helping moderators instantly generate, group, and answer audience questions while a session is live.
- Built a **live AI chatbot/assistant** for event audiences and a **post-event AI chatbot** that lets users ask questions about the presentation and content after the session ends, both powered by the same RAG system.
- Used the same AI system for **livestream summarization** and **AI-driven insight reporting**, turning raw session data into clear, readable summaries and engagement insights for clients.
- Built an **AI agent** (on AWS Bedrock) that lets moderators ask the AI directly about live user metrics in plain language, offering a faster, conversational alternative to traditional report generation.
- Created **Mission Control**, a unified moderator dashboard that brings live-event interaction, moderation, and reporting features together in one place.
- Built real-time audience interaction on **AWS AppSync, GraphQL, and DynamoDB** — Q&A (with upvoting, direct replies to attendees, and a teleprompter view for speakers), live polls, surveys, quizzes, and synchronized slideshows — replacing the older AWS IoT Core setup with a faster, more scalable solution across 50+ live events.
- Implemented **Amazon IVS** real-time, ultra-low-latency streaming, which broadcasts live video over WebRTC to keep audience and presenter in sync within fractions of a second, enabling true two-way interaction during live sessions.
- Added **live video transcription and translation** so audiences could follow sessions with real-time captions in their own language.
- Secured stored video and generated reports with encryption using **AWS KMS**, protecting sensitive client and event data at rest.
- Built a **QR-based check-in app with Flutter**, along with attendance reporting for on-site event registration.
- Built a **telemetry pipeline** to capture live-event activity, streaming events through **Amazon Kinesis Firehose** into **Amazon S3** for long-term storage while writing to **Redis / Valkey** for live statistics and real-time user information during the event.
- Used the stored S3 data with **AWS Glue** and **Athena** after each event to generate detailed reports and AI-driven insights.
- Built analytics workflows with **AWS Step Functions** to combine logins, Q&A, polls, and survey data across the pipeline without hitting Lambda timeouts.
- Migrated old Jenkins pipelines to **AWS CodeBuild** and set up multi-account, cross-region deployments using STS role assumption.
- Cut AWS costs by cleaning up unused infrastructure and adding cleanup rules for detached EBS volumes, idle Elastic IPs, and S3 lifecycle policies.
- Converted 1,000+ video files to HLS with AWS MediaConvert and built customized event webpages for 30+ clients using Nova's web builder (jQuery and Vue), with automated client domain setup and dynamic language localization.

Senior Full-Stack Developer / Team Lead

Jun 2024 – Dec 2024 · Remote

Bitsky

Online gaming and entertainment platform.

- Integrated **third-party game providers** (such as Panda 777) into the platform, handling game session management, callbacks, and real-time balance updates.
- Built a **bonus and points system** that rewards users based on their gaming performance and activity.

- Implemented a secure **Stripe** payment and wallet system for deposits, transactions, and payouts, with proper transaction handling and reconciliation.

Senior Full-Stack Developer / Web Team Lead

Jan 2022 – Dec 2023 · Lalitpur, Nepal

Swivt Technologies

IT company delivering products and solutions across multiple industries.

- Led a team of 5–8 developers to deliver 5+ projects, coordinating Jira and Trello workflows with the project manager to improve delivery.
- Built **Crystal Academy**, an online e-learning platform delivering courses with tests and certificates.
- Developed e-commerce platforms including a Taekwondo studio store and **Aitken Vanson**, a shoe retailer, with online booking and Stripe payment integration.
- Worked on a **food and grocery super app**, building features across ordering, delivery, and vendor management.
- Built a custom CMS to manage content across multiple services and platforms, making content updates much faster and easier.
- Hosted and managed 5+ client websites using Cloudflare and GoDaddy.

Full-Stack Developer

Dec 2018 – Jan 2022 · Kathmandu, Nepal

Upaya Business Solutions

Online marketplace for business registration, compliance, and financial services.

- Built a frontend search system that queries 1M+ records in 0.3–0.7 seconds using IndexedDB.
- Built tailored accounting systems for 5 departments together with the accounts and customer-relation teams, improving accounting efficiency.
- Developed a customer-relation system to improve how the company managed client relationships.
- Automated daily, weekly, and monthly finance and visit summaries to speed up reporting.
- Built 5 hybrid mobile apps using Cordova.

Full-Stack Developer

Jun 2018 – Feb 2019 · Kathmandu, Nepal

Upaya

Tech-driven logistics provider offering intra-city and inter-city delivery across Nepal.

- Worked in a 5-member team to build an end-to-end logistics management system.
- Added real-time shipment tracking with a live GPS feed, letting customers monitor deliveries as they happen.
- Set up a centralized customer service portal to improve order processing efficiency.
- Built a dynamic pricing algorithm based on route length, vehicle type, and traffic conditions.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, PHP, Python, Node.js

Frontend & Mobile: React.js, Vue.js, jQuery, Flutter, Cordova

Cloud & Serverless: AWS (Lambda, AppSync, Step Functions, Kinesis Firehose, Glue, Athena, S3, DynamoDB, EC2, CloudFront, IAM/STS, CloudWatch, IVS, MediaConvert, KMS), Serverless Framework, CloudFormation

AI: AI agents, RAG (Retrieval-Augmented Generation) systems, AI assistants/chatbots, knowledge bases & vector databases, semantic search, embeddings, LLM integration (Claude); AWS Bedrock

DevOps: AWS CodeBuild, Jenkins, CI/CD pipeline design, multi-account deployments, Git submodules, Bash scripting, Docker

Databases: PostgreSQL, DynamoDB, MongoDB, MySQL, Redis / Valkey, IndexedDB

API & Security: GraphQL (mutations, subscriptions), AWS Cognito, Lambda custom authorizers, multi-tier authorization

EDUCATION

Bachelor's Degree in Information Management · **Tribhuvan University**

2014 – 2018

SELECTED PROJECTS

Nova Dynamic Media — novaweb.live

Upaya — upaya.com.np

Upaya Business Solutions — ubs.com.np

Bitsky — bitsky.bet

WorkItPT — workitpt.goswivt.com

Crystal Academy — crystalacademy.org